

FNP (Ferns and Petals) Sales Analysis Report

Problem Statement

You have been given a dataset from FNP (Ferns and Petals) that specializes in sending gifts for various occasions like Diwali, Raksha Bandhan, Holi, Valentine's Day, Birthdays, and Anniversaries. The dataset contains details about the products, orders, customers, and relevant dates. The task is to analyze this dataset to uncover key insights related to sales trends, customer behavior, and product performance, and to help the company improve its sales strategy and optimize customer satisfaction.

Dataset Description

The analysis is based on 3 datasets:

- **customers.csv** — 204 records containing Customer_ID, Name, City, Gender, Contact Number, Email, and Address
- **orders.csv** — 1,000 records containing Order_ID, Customer_ID, Product_ID, Quantity, Order_Date, Order_Time, Delivery_Date, Delivery_Time, Location, and Occasion
- **products.csv** — 70 records containing Product_ID, Product_Name, Category, Price (INR), Occasion, and Description

These 3 tables were linked via Customer_ID and Product_ID in Power Pivot to build the data model.

Objective

The objective of this analysis is to answer the following 10 key business questions:

- Total Revenue — Identify the overall revenue
- Average Order and Delivery Time — Evaluate the time taken for orders to be delivered
- Monthly Sales Performance — Examine how sales fluctuate across the months of 2023
- Top Products by Revenue — Determine which products are the top revenue generators
- Customer Spending Analysis — Understand how much customers are spending on average
- Sales Performance by Top 5 Products — Track the sales performance of top 5 products
- Top 10 Cities by Number of Orders — Find out which cities are placing the highest number of orders
- Order Quantity vs. Delivery Time — Analyze if higher order quantities impact delivery times
- Revenue Comparison Between Occasions — Compare revenue generated across different occasions
- Product Popularity by Occasion — Identify which products are most popular during specific occasions

Key Insights

1. Total Revenue & Orders: FNP processed 1,000 orders in 2023, generating a total revenue of ₹35,20,984. The average customer spending per order stands at ₹3,521, reflecting a mid-to-premium positioning of FNP's product offerings.

2. Average Order and Delivery Time: The average time between order placement and delivery is 5.53 days. This consistent turnaround indicates reliable logistics and operational efficiency across all order types and regions.

3. Monthly Sales Performance: Revenue peaks in February, driven by Valentine's Day demand, and again around August due to Raksha Bandhan. A significant dip is observed in mid-year months (May–June), indicating a seasonal slump during non-occasion periods.

4. Top Products by Revenue: The top 5 revenue-generating products are Magnum Set, Quia Gift, Harum Pack, Dolores Gift, and Deserunt Box — each contributing close to or above ₹1,00,000 in revenue.

5. Customer Spending Analysis: With an average spend of ₹3,521 per order across 204 unique customers, customers show a consistent willingness to invest in gifting. This presents an opportunity to upsell premium bundles and occasion-specific gift sets.

6. Sales Performance by Top 5 Products: The top 5 products maintain relatively balanced revenue contributions, with Magnum Set and Quia Gift slightly ahead. This suggests no single product dominates, indicating a healthy and diverse product mix across 70 total products.

7. Top 10 Cities by Number of Orders: The highest number of orders come from:

- Kavali
- Imphal
- Dhanbad
- Bidhanagar
- Bilaspur
- Guntakal
- Haridwar
- North Dumdum
- Kaveri
- Bhagpara

These cities represent key geographic markets for FNP across India.

8. Order Quantity vs. Delivery Time: Delivery time remains stable at approximately 5.53 days regardless of order volume, suggesting FNP's fulfillment operations scale effectively during high-demand periods without impacting delivery speed.

9. Revenue Comparison Between Occasions: Anniversary and Holi are the top revenue-generating occasions, followed closely by Raksha Bandhan and Valentine's Day. Birthday and

Diwali contribute moderately, while All Occasions (non-specific) also shows significant volume.

10. Product Popularity by Occasion: Cakes dominate overall category revenue at approximately ₹10,00,000, followed by Soft Toys and Sweets. Colors perform strongly during Holi, and Raksha Bandhan drives demand for specific gift sets, showing clear occasion-product alignment across 7 product categories.

Recommendations

- Focus marketing campaigns around peak occasions — Anniversary, Holi, Raksha Bandhan, and Valentine's Day
- Launch mid-year promotions in May–June to address the seasonal revenue slump
- Prioritize stock availability for top products like Magnum Set and Quia Gift
- Expand delivery reach and run targeted promotions in high-order cities like Kavali, Imphal, and Dhanbad
- Introduce occasion-specific product bundles to increase average order value
- Highlight the 5.53-day delivery promise in marketing to build customer trust
- Leverage customer data (204 profiles) for personalised email and retargeting campaigns

Conclusion

The analysis of FNP's 2023 data — spanning 1,000 orders, 204 customers, and 70 products — reveals a business that performs strongest during key gifting occasions. Cakes, Soft Toys, and Sweets are the most popular categories, while Anniversary and Holi lead occasion-based revenue. Cities like Kavali and Imphal show strong regional demand. With consistent delivery performance and a healthy average spend of ₹3,521, FNP is well-positioned to grow further by targeting peak seasons, high-demand cities, and popular product categories more aggressively in the coming year.